Physical Reachability Limits of Pattern Inverse-Design for Centrally-Driven Chladni Plates:

A Symmetry-Constrained, Differentiable-Surrogate-to-FEA Framework with Manufacturable Output

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Abstract

We study a classic and severely constrained inverse problem: given a target image, recover a plate (a spatial thickness field, and optionally a per-cell fibre orientation) together with a set of drive frequencies whose Chladni sand pattern approximates the target. The problem is posed under a fixed, non-negotiable hardware contract: a 150 mm square plate discretised by a 15×15 thickness grid, an 8 mm central clamp, free edges, and a *single* central point excitation with a frequency sweep, with a high-fidelity finite-element model (COMSOL Multiphysics with LiveLink™ for MATLAB) as the primary source of truth. Across an extended development effort spanning three algorithmic "eras", roughly twenty algorithm lines and several hundred finite-element candidates, we converge on four central, and largely counter-intuitive, conclusions. (i) Arbitrary custom-pattern inversion is *physically impossible* rather than an engineering shortfall: the dihedral symmetry group D_4 of the configuration forces a central point drive to couple only into the A_1 irreducible representation, so the fraction of target energy lying in A_1 is a hard reachability ceiling. (ii) Material anisotropy (the fibre orientation θ) is essentially inert in the working band, because the inertial term dominates the stiffness term by several orders of magnitude and θ enters the physics only through stiffness. (iii) The gap between a differentiable thin-plate surrogate and the thick-plate finite-element model is the dominant practical ceiling, and is a systematic physical difference rather than numerical noise. (iv) The honest deliverable is therefore not "arbitrary inversion" but a reproducible, manufacturable *system*: a differentiable surrogate used as a compass, a two-phase finite-element production pipeline (modal selection followed by trust-region refinement), and a directly 3D-printable export. We develop the theory and the methodology in full; the quantitative experimental validation is reported separately and the corresponding section here is left as a scaffold.

Keywords: Chladni figures; inverse design; structural dynamics; plate vibration; dihedral symmetry; irreducible representations; differentiable surrogate; topology/thickness optimisation; finite-element validation; additive manufacturing.

1 Introduction

1.1 Background and motivation

Chladni figures are the patterns traced by fine powder on a vibrating plate: the powder migrates away from antinodes and accumulates along nodal lines where the out-of-plane displacement is minimal, rendering an eigenmode visible to the naked eye [1, 2]. The *forward* problem—predicting the nodal pattern from the plate geometry, material and drive frequency—is, in principle, a standard exercise in linear structural dynamics. The *inverse* problem we consider here is far harder: *given a desired target image, recover a plate design and a drive frequency whose Chladni* *pattern reproduces it.* Inverse design of this kind sits at the intersection of structural-dynamics eigenvalue placement, shape/topology optimisation, and manufacturability, and it is an attractive vehicle for studying how far a purely computational pipeline can be pushed against a genuinely fixed piece of hardware.

1.2 Problem statement

We fix a square plate excited at its centre by a single point source and ask whether, by spatially varying the plate thickness (and, as a tempting extra degree of freedom, the local fibre orientation of a 3D-printed part) and by choosing one or more sweep frequencies, the resulting nodal pattern can be made to resemble an arbitrary target such as a letter pair, a cross, a diagonal, or a ring. The constraints (Section 3) are immutable for the entire study; every algorithm must be designed *around* them rather than relaxing them.

1.3 Contributions and central thesis

Our central thesis is that the most valuable output of such a study is not a demonstration that "any pattern can be produced"—which we show does not exist physically—but a precise characterisation of *where the reachability boundary lies, why it lies there, and how to engineer the boundary-hugging optimum into a deliverable product.* Concretely, this paper contributes:

(1) a **symmetry-based reachability theory** that identifies the A_1 irreducible representation of the dihedral group D_4 as the only channel a central point drive can excite, yielding a closed-form, target-specific upper bound on what is achievable (Section 4);
(2) a **scaling analysis of anisotropy** showing that the fibre orientation θ is a parasitic degree of freedom in the working band, with a mass–stiffness scale separation of several orders of magnitude that renders its sensitivity negligible (Section 5.2);
(3) a **differentiable orthotropic-plate surrogate** with adjoint/autograd sensitivities, framed explicitly as a "compass, not a judge" (Section 5.1);
(4) a **recognisability metric** grounded in the actual powder-deposition physics rather than pixel overlap (Section 5.3);
(5) a **two-phase finite-element production pipeline** (modal selection followed by trust-region refinement) that manages the surrogate–FEA gap explicitly (Section 5.4); and
(6) a **manufacturable export and software system** that turns a design into a printable part (Section 5.6).

Throughout, we adopt a deliberately honest, project-report register: negative results, retracted assumptions, and methodological mistakes are reported as first-class findings, because in a fixed-hardware inverse problem they delimit the feasible region as sharply as any positive result.

1.4 Organisation

Section 2 reviews directly comparable prior art. Section 3 formalises the problem and the hardware contract. Section 4 develops the symmetry ceiling. Section 5 presents the full methodology (surrogate, anisotropy analysis, metric, pipeline, export, software). Section 7 is the experimental section and is intentionally left as a scaffold to be completed. Section 8 discusses the lessons; Section 9 states the limitations and future work; Section 10 concludes.

2 Related work

Topology optimisation of Chladni / modal patterns. The most directly comparable prior art is the 2024 study on topology-optimisation design of Chladni patterns for vibration-mode manipulability [45], which uses density-based topology optimisation (SIMP) with an objective that minimises the error between computed eigenvectors and a target Chladni pattern, inversely designing material distributions for single- and multi-order patterns. Its eigenvector-error objective is cognate to our own scoring evolution, and its SIMP density variable is close in spirit to our density/thickness field. The key difference is that it is *not* bound by a single-central-excitation D_4 constraint—the very source of our A_1 ceiling—so its reachable pattern space is intrinsically larger. The methodological lineage of mode-targeting SIMP traces back to topology optimisation of resonating structures [46] and, more broadly, to eigenvalue topology optimisation [37, 38] within the classical density-based framework [21, 22, 16].

Energy-landscape formulations. An open-source line of work frames Chladni inverse design as an "energy-landscape" problem in which a nonnegative mixture of squared mode shapes, $E(x, y) = \sum_k b_k \, \varphi_k(x, y)^2$, is shaped toward the target [47]. This matches our amplitude-valley loss and independently confirms a structural bias: nonnegative squared mixtures prefer boxes, bands, grids, crosses and large connected glyphs, and resist fine, separated features—a bias strongly consistent with what our symmetry analysis and powder physics predict.

Cymatics and wave control in structured plates. Reshaping flexural-wave fields and nodal lines through plate microstructure has been demonstrated in the context of flexural cloaking [48], and forced-response Chladni patterns have been related to maximum-entropy states of an inhomogeneous Helmholtz equation [49], both of which inform the distinction between forced response and free vibration that we revisit in Section 5.5.

Hardware-driven control. Finally, multi-actuator active control of particles on a Chladni plate [50] demonstrates that genuinely breaking the symmetry ceiling requires acting on the *hardware*—additional shakers or off-centre excitation—a conclusion that our theory reaches from the opposite direction and that frames our recommendations for future work. Level-set shape optimisation [51] is noted as an alternative geometry representation that we did not adopt, preferring a density/thickness field.

3 Problem formulation and hardware contract

3.1 Forward model

Let $\Omega = [0, L]^2$ be the mid-surface of a thin plate of spatially varying thickness $h(\boldsymbol{x})$, clamped on a small central disc and free on all four edges. Under the Kirchhoff–Love thin-plate assumption, the free-vibration eigenproblem is the biharmonic-type generalised problem

$$\mathbf{K}(\boldsymbol{h}) \, \boldsymbol{\phi}_i \;=\; \omega_i^2 \, \mathbf{M}(\boldsymbol{h}) \, \boldsymbol{\phi}_i, \tag{1}$$

where $\mathbf{K}$ and $\mathbf{M}$ are the (discretised) stiffness and mass operators, ω_i the angular eigenfrequencies and $\boldsymbol{\phi}_i$ the mode shapes. A high-fidelity description retains transverse shear via the Mindlin–Reissner thick-plate theory [3, 4], which is the model used inside the finite-element solver. Under a steady single-point harmonic excitation of angular frequency ω and load vector $\boldsymbol{f}$, the forced response is

$$\boldsymbol{u}(\omega) = \left(\mathbf{K} - \omega^2 \mathbf{M} + \mathrm{i}\, \omega \, \mathbf{C}\right)^{-1} \boldsymbol{f}, \tag{2}$$

with $\mathbf{C}$ a damping operator. The Chladni pattern is the locus of small $|\boldsymbol{u}|$: powder is deposited where the out-of-plane amplitude is minimal, which we model by a powder-density functional that concentrates mass on nodal lines (Section 5.3).

3.2 Inverse problem

The inverse problem seeks design variables that make the deposited pattern resemble a target image T. The primary design variable is the thickness field, discretised on the mandated grid,

$$\mathbf{H} = \{h_{mn}\}_{m,n=1}^{15} \in \mathbb{R}^{15\times 15}, \tag{3}$$

optionally augmented by a per-cell fibre orientation field $\boldsymbol{\theta}$ and by a small set of K drive frequencies $\{\omega_k\}$ with weights $\{w_k\}$. The objective is a recognisability functional $\mathcal{R}(\,\cdot\,; T)$ defined in Section 5.3.

3.3 Immutable hardware contract

Table 1 lists the constraints that remained fixed throughout the study and that define the physical ceiling of the problem.

Table 1: Immutable hardware contract. Every algorithm is designed around these.

Item	Value	Source
Design grid	15×15 thickness field	FEA `H.csv` contract
Plate size	150 mm $\times$ 150 mm	one-shot print boundary
Central clamp	radius 8 mm disc	standard rig
Edges	free on all four sides	free vibration
Excitation	single central point + sweep	no multi-shaker / added mass
Primary feedback	FEA eigenfrequency / forced response	LiveLink with MATLAB
Thickness range	0.6–2.0 mm, neighbour $\Delta \leq 1.0$ mm	manufacturability

4 The symmetry ceiling

4.1 Dihedral symmetry of the configuration

A square plate with a central clamp and a central point drive is invariant under the dihedral group D_4 of order eight (four rotations and four reflections). Because both the operator pencil $(\mathbf{K}, \mathbf{M})$ and the load $\boldsymbol{f}$ of a *centred* excitation are D_4-symmetric, the entire steady-state response must transform according to the representation theory of D_4 [7]. The group has five irreducible representations (irreps): the one-dimensional A_1, A_2, B_1, B_2 and the two-dimensional E.

4.2 Central excitation couples only to A_1

The load of a point force applied exactly at the (clamped) centre is itself invariant under every element of D_4; it therefore lies entirely in the trivial representation A_1. By orthogonality of irreps, such a load can only excite response components that also transform as A_1: the projection of the forced response onto any non-A_1 subspace vanishes identically. We implement this projection explicitly as a D_4 irreducible-representation decomposition of any candidate field. The practical consequence is a clean, *closed-form* reachability statement:

$$\boxed{\text{reachability}(T) \;=\; \frac{\|\mathcal{P}_{A_1} T\|^2}{\|T\|^2}\,,} \tag{4}$$

where $\mathcal{P}_{A_1}$ projects the target onto the A_1 subspace. Energy of the target outside A_1 cannot be reproduced *no matter how strong the algorithm*, because the central drive provides no channel into those irreps.

4.3 Worked reachability of representative targets

Equation (4) is analytic and can be evaluated directly on each target. Table 2 reports the resulting A_1 fractions. An X-shape lives entirely in A_1 and is in principle fully reachable; a single diagonal is only half-reachable (its complement lies in B_2); and the "IC" letter pair places the majority of its energy in E and B_1, so most of its signal is *structurally* inadmissible. This furnishes, for the first time in the project, an exact mathematical answer to the question "why does IC never appear" that is independent of any particular optimiser.

Table 2: Analytic reachability (Eq. (4)): the A_1 energy fraction is a hard upper bound a central drive can hope to reproduce.

Target	A_1 fraction	Outside A_1 (unreachable)	Meaning
X-shape	100.0%	0%	theoretically fully reachable
Single diagonal	51.7%	48.3% (B_2)	half reachable
IC letters	41.9%	58.1% (E+B_1)	majority cannot couple in

The implication for the whole pipeline is sharp: a target must first be *admissible* (high A_1 fraction) before any amount of optimisation is worthwhile. Using the reachability verdict as a front-end filter—rather than discovering the ceiling empirically after the fact—is, in retrospect, the single most important methodological lesson of the project.

5 Methodology

5.1 Differentiable orthotropic-plate surrogate

At the heart of the pipeline is a differentiable Kirchhoff plate surrogate implemented so that the loss is end-to-end differentiable with respect to the thickness field. We assemble the orthotropic bending-stiffness operator from the full laminate energy functional [6]; each cell may, in the general formulation, rotate its bending-stiffness tensor by a local angle θ. The discrete eigenproblem (1) is solved with a sparse solver, and gradients of the loss with respect to $\mathbf{H}$ (and $\boldsymbol{\theta}$, ω) are obtained by a combination of automatic differentiation [30, 31] and analytic eigenpair sensitivities [32, 33], the latter being the classical route to differentiating eigenvalues and eigenvectors of (1). The adjoint viewpoint [29] makes the gradient computation independent of the number of design variables.

The decisive design decision is one of *role*, not of formulation: the surrogate is treated as a **compass, not a judge**. It is fast enough to push the thickness field and frequency to a "surrogate-good" region in seconds, but it is never allowed to decide the winner; that authority belongs to the finite-element model. Two opposite misuses—treating the surrogate as the truth, and treating it as the final referee—were both committed earlier in the project and both produced over-optimistic designs that did not survive high-fidelity re-evaluation (Section 5.5).

5.2 The anisotropy hypothesis and its collapse

A tempting hypothesis motivated the orthotropic formulation: since 3D-printed parts are naturally anisotropic, letting the fibre orientation θ vary per cell would make the bending-stiffness operator cease to commute with D_4, allowing a central drive to leak into non-A_1 irreps and thereby break the ceiling of Section 4. The argument is physically appealing.

A scaling analysis nonetheless predicts that the effect is negligible in the working band. Writing the forced operator as $\mathbf{K} - \omega^2\mathbf{M}$, in the relevant frequency range the inertial term $\omega^2\|\mathbf{M}\|$ exceeds the stiffness term $\|\mathbf{K}\|$ by several orders of magnitude. Because θ enters the physics *only* through $\mathbf{K}$, its influence on the response is suppressed by the same ratio: the response is, to leading order, mass-controlled, and the symmetry-breaking that θ would introduce through the stiffness is drowned out. The orientation field is, in addition, a *parasitic* optimisation dimension:

it inflates a non-convex loss landscape and diverts gradient budget away from a clean convergence in $\mathbf{H}$ and ω. Independent confirmation that the realised anisotropy ratio of representative printable materials is far smaller than the optimistic literature values only reinforces the conclusion. We therefore retain the orthotropic machinery for generality but treat θ as effectively frozen, and the isotropic path as the default. The corresponding quantitative ablation is reported in Section 7.

5.3 A recognisability metric grounded in powder physics

Early scoring used pixel-overlap measures (IoU/Jaccard, Dice, Chamfer distance) [18, 19, 20], which are discrete, non-differentiable, and unstable under mode switching. We replace them with a metric grounded in the actual deposition physics: powder accumulates where $|\boldsymbol{u}| \approx 0$, so the relevant quantity is a powder density concentrated on nodal lines. The recognisability functional combines (i) *enrichment*, the ratio of powder density inside the target region to the global mean; (ii) *recall*, the fraction of the target covered; (iii) *contrast*; and (iv) a *direction* term. A continuous, differentiable surrogate of this functional—an amplitude-valley loss—is used inside the optimiser, while the discrete powder model is used for evaluation. We also note the connection to optimal-transport objectives: matching nodal-line measures rather than pixels is naturally expressed through Sinkhorn divergences [40], which, like Wasserstein objectives for waveform matching [41], avoid the cycle-skipping pathologies of pixelwise L^2/IoU losses. The choice of metric is not cosmetic: it re-shuffles which candidates are considered "best" and therefore steers the entire search, a point we return to in Section 8.

5.4 Two-phase finite-element production pipeline

The production pipeline couples the surrogate to the finite-element model in two phases that decouple the two things that can go wrong—picking the wrong frequency and picking the wrong shape.

Phase 1 modal selection. We compute the finite-element eigenfrequencies of the candidate plate, rank each eigenmode by its likeness to the target, drive the forced response (2) *on resonance*, and compose the best single mode or a composite of complementary modes. Driving on resonance is essential: an off-resonance drive lets the central point force dominate and skews the modes, so that clean nodal lines never form. We also find, in principle, that an *interference of two complementary modes* (a beat composite) can be more target-like than any single eigenmode—a degree of freedom invisible to a single-mode view.

Phase 2 trust-region refinement. Taking the finite-element response as ground truth, the thickness field is refined by a trust-region iteration [42] embedded in a surrogate-management framework [43]: the surrogate proposes a step, the finite-element model re-evaluates it, and the step is accepted or the trust radius shrunk. Modal correspondence between iterations is tracked by the Modal Assurance Criterion [44]. This explicitly manages the surrogate–FEA gap through "trust" and refines monotonically, in contrast to a naive loop that corrects frequency and shape simultaneously and diverges.

5.5 The surrogate–FEA gap

The thin-plate surrogate and the thick-plate finite-element model differ permanently: plate theory (thin versus shear-deformable thick), mesh density, boundary geometry (an abstract central constraint versus a circular clamp), and two-dimensional plane stress versus a full three-dimensional elasticity tensor. These are *systematic physical differences*, not noise. Qualitatively, the surrogate is excellent at low modal order and degrades at high order, and the mapping between surrogate and finite-element frequencies follows a clean power law rather than a random scatter—evidence that the gap is structural. The correct response to a systematic gap is neither to trust the surrogate nor to over-fit against it, but to use it as a compass and let the finite-element model arbitrate; this is precisely the rationale for the two-phase pipeline of Section 5.4. A separate, sobering observation concerns numerical hygiene: a single cross-scale additive constant in a normalisation ($\mathrm{amp}/(\max(\mathrm{amp}) + \varepsilon)$ with a fixed ε) silently clamped a key metric whenever the finite-element amplitudes were many orders of magnitude smaller than the surrogate's, systematically *under*-reporting historical results until corrected.

5.6 Manufacturable export and software system

The pipeline terminates in a manufacturable artefact. A zero-dependency exporter converts the 15×15 thickness field into a stepped-box STL (one box per cell, flat-bottomed for bed adhesion, with optional smoothing of the steps), directly slicable for fused-deposition printing without any orientation control. The system is delivered as a full-stack application: a multithreaded backend that orchestrates the surrogate and the finite-element solver (managing the solver lifecycle, auto-discovering tool paths, and exposing result and diagnostic endpoints including the reachability verdict of Section 4), and a single-page frontend organised around target design, material tuning, run control, results inspection, a uniform-plate pattern gallery filtered by central-drive excitability, and a 3D-print export tab. The engineering discipline of the project included a read-only algorithmic audit that quantified cognitive biases in the metrics, strict roll-back discipline against regressions, and per-experiment reproducibility.

6 Algorithmic evolution: three eras and roughly twenty lines

The architecture of Section 5 did not arrive by design; it precipitated out of an extended trial-and-error process that is itself part of the contribution, because each abandoned line marks a wall of the feasible region. We organise the history into three eras followed by a modern production period (Table 3). The milestone numbers quoted below describe the *development trajectory*; the controlled experimental validation remains the (deliberately empty) subject of Section 7.

Table 3: The three eras and the modern production period.

Era	Phases	Core paradigm	Outcome
I — Blind search	Phase A–I (9 lines)	"guess thickness → run FEA → tweak"	score stuck at 0.034; all below bar
II — Structured	W1–W6 (6 lines)	prior filter + main-loss redesign + gradient path	IC score pushed to 0.171
III — Recognisability	W7–W8 (2 lines)	redefine success via powder physics	X-shape enrichment 3.76× (ceiling)
Modern production	W9, W10, two-phase (∼3 lines)	differentiable surrogate + FEA trust-region loop + productisation	two-phase pipeline + software

6.1 Era I — blind search (Phases A–I)

The most naive formulation treats the 225-dimensional thickness field directly as the optimisation variable and searches it with gradient-free methods (Table 4). Random and evolutionary search only ever produced the "generic plate-mode family" (arcs, stars, rings, centre-dominant blobs) with no letter-like structure; response-guided error maps could say *where* a pattern was wrong but not *which* nodal line to move where, because eigenmodes are global objects; a first Kirchhoff–Love proxy already exposed the surrogate-to-FEA gap; and a strict-scoring upgrade honestly cut an inflated best of ≈ 0.2 down to 0.034. Expanding the design contract from 225 to 675 dimensions did not help. Era I ended with a best honest score of 0.034, far below the 0.08 engineering bar, and a clear diagnosis: more generations would not help; the problem had to be redesigned from the physics and mathematics upward.

Table 4: Era I — the nine blind-search lines and their lessons.

Line	Method	Methodological basis	Lesson
A	Random / evolutionary search (GA, mutation, neighbour-diff repair, skeleton guidance)	[15, 16]	Only the generic plate-mode family; no letters
B	Response-guided loop (missing/extra error maps; iterative inversion)	[17]	Error maps locate error but not the corrective move; modes are global
C	Kirchhoff–Love plate proxy (sparse biharmonic eigensolve)	[5, 2]	First exposure of the surrogate-to-FEA gap; proxy scores do not transfer
D	Strict shape scoring ($v1 \to v5$; IoU/Dice/Chamfer)	[18, 19, 20]	Honest metrics beat pretty ones: best falls from ≈ 0.2 to 0.034
E	Design-contract expansion (SIMP density/loss/material, $225 \to 675$)	[21, 22]	More dimensions are not the answer; placement still fails
F	Auxiliary-physics local heuristics (direct/pattern search)	[23]	Most robust historical source, but capped at ≈ 0.032–0.034
G	Latent joint physics (hand-designed basis; POD)	[24]	A hand-made basis can miss key directions
H	CMA-ES in latent coefficient space	[25]	Exposes surrogate optimism: good on proxy, poor on FEA
I	Response-calibrated scheduling (surrogate/Bayesian optimisation)	[26, 27]	First systemic treatment of "surrogate is unreliable"; not fully landed

6.2 Era II — structured search (W1–W6)

The second era replaced blind search with structure: a physical prior filter, a differentiable main loss, a connected gradient path, and a data-driven design subspace (Table 5). Two points deserve emphasis. First, the realizability prior of W1 already *predicted the entire IC outcome* through a D_4-mismatch of 0.78; that this verdict was not used as a front-end filter at the time is, in retrospect, the single biggest methodological miss of the project (it is precisely the theory later formalised in Section 4). Second, W6 — the climax of the era — pushed the IC score from 0.034 to 0.171, a roughly fivefold gain that nonetheless still did not look like IC, which is exactly what motivated the redefinition of success in Era III.

6.3 Era III — recognisability (W7–W8)

Era III was triggered by the realisation that a good score (0.171) still did not look like the target. Success was redefined to accept "similar, not identical", and the scoring system was rewritten around powder physics. The multi-frequency root-mean-square composition of W7, optimised with a first-order stochastic method [39] over the forced response [5], proved a *pseudo*-breakthrough: the optimiser learns to weight $|\boldsymbol{u}|^2$ to fabricate low-amplitude regions that the finite-element model cannot reproduce, and it tends to collapse back to a single frequency. W8 then replaced pixel overlap with the powder-deposition metric of Section 5.3, formulated through an optimal-transport view of nodal-line matching [1, 40, 41]. Re-ranking the entire candidate history under the new metric revealed that the project had optimised the *wrong* metric for months and that better candidates had been buried. The X-shape enrichment reached 2.84×, and 3.76× after the normalisation fix discussed below, which became the project ceiling.

Table 5: Era II — the six structured lines and their key results.

Line	Method	Methodological basis	Key result / lesson
W1	Realizability prior filter (Courant nodal domains + D_4 mismatch + stroke width)	[8, 9, 2]	IC mismatch 0.78 predicted the ending; not used up front (biggest miss)
W2	Main-loss redesign (IoU $\rightarrow$ amplitude valley)	[28]	Smoother optimisation; physical ceiling unchanged
W3	Differentiable plate + adjoint ($\partial\,\mathrm{loss}/\partial\mathbf{H}$)	[29, 30, 31, 32, 33]	Gradient path finally connected; first structural upgrade
W4	Low-dimensional manifold search (PCA of candidates)	[34, 24]	Data basis beats hand-made basis; markedly lower loss than W3
W5	Homotopy continuation (warm-start toward target)	[35, 36]	Stalls at $\lambda \approx 0.3$ — an honest reachability indicator
W6	Spectral placement (cluster eigenvalues near drive)	[37, 38]	Era II climax: IC score 0.034 $\rightarrow$ 0.171 (5$\times$), still not legible

6.4 Modern production — W9, W10 and the two-phase pipeline

The modern period built the pieces that the earlier eras had only gestured at. The W9 trust-region loop was structurally correct but under-tuned (too few inner steps, too-weak correction, a trust radius that shrank to 0.094 mm), and its best finite-element result of 1.40$\times$ did not break through — the lesson being that circling a poor surrogate does not help when the root cause is the surrogate physics itself; W9 nonetheless contributed the trust-region and surrogate-management machinery with modal tracking [42, 43, 44] that Phase 2 later inherited. W10 implemented the differentiable orthotropic plate of Section 5.1 [6], and the two-phase pipeline of Section 5.4 finally realised the iterative calibration loop that Era I had owed all along, by decoupling the frequency correction (driven directly at the finite-element resonance) from the shape correction (a small trust-region residual).

6.5 Cross-cutting empirical findings

Three findings recur across the eras and deserve separate statement.

On-resonance versus off-resonance drive. A late, bug-level discovery: Phase 1 once defaulted to driving at $f_{\mathrm{eig}} + 1.5\,\mathrm{Hz}$ (off-resonance) to avoid a solver singularity, but this let the central point force dominate the response and skewed the modes, so that cross- and X-patterns never formed. Switching to an on-resonance drive (offset zero) instantly recovered clean modal nodal lines. The off-resonance artefact also explains a spurious "bar through the middle": off resonance, the centre is an antinode bright streak pushed up by the point force, not a true nodal line.

A cross-scale numerical constant once hid the truth. The powder-density normalisation $\mathrm{amp}/(\max(\mathrm{amp}) + \varepsilon)$ with a fixed ε was harmless in surrogate-space, where amplitudes are $\mathcal{O}(1)$, but in finite-element space, where amplitudes are $\mathcal{O}(10^{-12})$, the constant was a thousand times larger than the signal and silently clamped the key metric to ≈ 1.00. After the fix, all historical finite-element numbers rose (the X-shape from 2.84$\times$ to 3.76$\times$). The lesson is blunt: a cross-scale additive constant is a silent killer.

The metric defines the direction of search. Five primary metrics were used over the project, and each switch re-shuffled the "best candidate" (one W6 variant was first under one metric and tenth under another): pixel overlaps (IoU/Dice/Chamfer); a smoothed pixel score;

an amplitude-valley loss; a multi-frequency RMS (easy for the optimiser to cheat); and finally the powder-physics recognisability of Section 5.3. A separate read-only algorithmic audit later quantified residual biases even in the last family (for example, that enrichment can prefer a compact central blob over an extended shape, and that fixed-percentile thresholds are sensitive to powder density). The meta-lesson, which we restate in Section 8, is to define "success" before optimising it.

7 Experimental setup and results

This section is intentionally left as a scaffold; the quantitative experimental write-up is pending and will populate the subsections below. No numerical results are reported here yet.

7.1 Experimental setup

[To be completed.] Plate and material parameters, the finite-element model configuration (mesh, element type, damping, eigenfrequency and frequency-domain solver settings via LiveLink), the powder model parameters, optimiser hyper-parameters, and the target set.

7.2 Validation of the reachability theory

[To be completed.] Empirical confirmation that achievable recognisability tracks the analytic A_1 fraction of Table 2 across the target set.

7.3 Anisotropy ablation (material battery)

[To be completed.] A target $\times$ material ablation comparing an anisotropic configuration (orientation active) against an isotropic baseline (orientation frozen), together with the direct sensitivity measurement of the composite amplitude to θ.

7.4 Quantification of the surrogate–FEA gap

[To be completed.] Modal-assurance statistics across modal order, the surrogate-to-finite-element frequency map, and the fraction of the surrogate-predicted improvement that the finite-element pipeline reproduces.

7.5 End-to-end case studies

[To be completed.] Full pipeline results for representative targets (X-shape, IC, cross, single diagonal, ring), including the best single-mode and composite drives, the refined thickness field, and the exported printable geometry.

8 Discussion

The four findings of the abstract cohere into a single design philosophy. First, the reachability theory of Section 4 reframes failure: when a target such as IC cannot be produced, this is not an algorithmic shortfall but a structural fact about which irreps a central drive can excite, and it is better to state that fact precisely than to keep optimising against an inadmissible target. Second, the anisotropy analysis of Section 5.2 is a cautionary tale about "physically plausible" degrees of freedom: a mechanism can be real (orientation does break symmetry through the stiffness) and yet be quantitatively irrelevant (because the response is mass-controlled in the working band), so a favourite hypothesis must be allowed to die when the scaling says so. Third, the surrogate–FEA gap of Section 5.5 shows that the operative ceiling in practice is often not the deep physics but the fidelity gap between the model one optimises and the model one believes, and that the disciplined response is to assign each model the role it is good at. Fourth, the metric discussion of

Section 5.3 underlines that in an inverse problem the definition of "success" is itself a design choice that silently steers the search; defining it correctly—here, in terms of powder physics rather than pixel overlap—is prerequisite to optimising at all. The recurring meta-lesson is that a substantial fraction of effort in a fixed-hardware inverse problem is well spent on *characterising the boundary* rather than on pushing against it.

9 Limitations and future work

The limitations are inherent rather than incidental. Any target that is not D_4-compatible—an arbitrary letter pair, a single diagonal, an L-shape, a generic logo—is physically unreachable under a central point drive, and this is a hard ceiling rather than a tuning problem. Material anisotropy offers little headroom, both because realised anisotropy ratios are modest and because the orientation gradient is negligible off resonance. A surrogate is never the finite-element model, so any surrogate-only "breakthrough" must be re-checked at high fidelity.

It follows that genuine further progress requires changing the *hardware*, in decreasing order of leverage: a D_4-symmetry-breaking excitation (multiple shakers or an off-centre drive), a changed boundary (a non-square plate), or discrete topology (holes and slots). These lie outside the fixed contract of the present study but are the only routes that can lift the A_1 ceiling. Within the present contract, promising refinements include a symmetry-aware direction metric (to fix principal-axis degeneracy on symmetric targets), a genuinely physical time-division multi-frequency drive rather than a numerical root-mean-square composite, and further tuning of the trust-region parameters of Phase 2.

10 Conclusion

Under a fixed hardware contract—a square plate with a single central excitation—we have shown that the right question is not "how strong can the inversion be" but "where is the reachability boundary, and why". A symmetry-based theory identifies the A_1 irreducible representation as the only channel a central drive can excite and turns this into a closed-form, target-specific ceiling; a scaling analysis demotes material anisotropy from a hoped-for escape route to a parasitic degree of freedom; and a two-phase, surrogate-as-compass pipeline manages the dominant practical obstacle, the gap between a thin-plate surrogate and a thick-plate finite-element model. The deliverable is an honest, reproducible, manufacturable system rather than an impossible promise of arbitrary inversion. The remaining headroom, we argue, is not in the algorithm but in the hardware. The experimental validation that quantifies these claims is reported separately; the corresponding section here is left as a scaffold by design.

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